A-A-51178
February 7, 1990

COMMERCIAL ITEM DESCRIPTION

ANODES, NICKEL, ELECTRO-REFINED

The General Services Administration has authorized the use of this commercial item description in preference to Type III, Styles A, B, and C of A-A-51142.

Abstract. This description covers electro-refined nickel anodes used in electroplating baths.

Classification. The anodes shall be of the following styles and sizes. The style and size shall be as specified.

Style A - Flat bar cut from electrolytic sheet.
Size 1 - 4 inches wide by 1/2 inch thick by 12 inches in length.
Size 2 - 4-1/2 inches wide by 1/2 inch thick by 28-1/2 inches in length.
Size 3 - 9 inches wide by 1/2 inch thick by 28-1/2 inches in length.
Size 4 - 12 inches wide by 1/2 inch thick by 28-1/2 inches in length.

Style B - Button-electrolytic.
Size 1 - 1 inch diameter by 1/4 to 3/4 inch in thickness.

Style C - Square.
Size 1 - 1 inch by 1 inch by 1/4 inch.

Salient characteristics. The anodes shall meet the following minimum requirements.

1. Chemical composition. The chemical composition of the anodes shall meet the requirements of table I.

Beneficial comments recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: Defense Industrial Plant Equipment Center, ATTN: DIPIC-SSG, 2163 Airways Blvd., Memphis TN 38114-5051.

AMSC N/A FSC 3426

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:52Z
Check the source to verify that this is the current version before use.

TABLE I. Chemical composition.

Element	Percent
Nickel plus cobalt 1/	99.85 min.
Iron	0.02 max.
Manganese	0.005 max.
Copper	0.02 max.
Silicon	0.005 max.
Sulfur	0.03 max.
Carbon	0.03 max.
Total other elements 2/	0.20 max.

1/ Cobalt counts as nickel.
2/ Other elements, such as P, As, Sb, Pb, and zinc individually determined, shall not exceed 0.005 percent.

2. Suspension. The method of suspension shall be in accordance with the manufacturers standard practice.

3. Workmanship. The anodes shall be clean and substantially free from cracks, warps, inclusions, porosity, surface film, or other defects which may adversely affect uniform corrosion in service.

Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this commercial item description, and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest revision of FED-STD-376, and all other requirements of this commercial item description are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request shall be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Regulatory requirements. In accordance with section 23.403 of the Federal Acquisition Regulations, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing suppliers' employees to undue hazards from the recovered materials.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:52Z
Check the source to verify that this is the current version before use.

Environmental protection. During the manufacture of the anodes, materials hazardous to the ecological system as prohibited by the Federal, state, or local statutes in effect on the date of the contract shall not be used or emitted.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection, examination, and test requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections, examinations, or tests set forth in this description where such inspections, examinations, and tests are deemed necessary to assure supplies and services conform to prescribed requirements.

Responsibility for compliance. All items shall meet all requirements of salient characteristics paragraphs. The inspection set forth in this description shall become a part of the contractors overall inspection system or quality program. The absence of any inspection requirements in the description shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

Sampling inspection. As a minimum sampling inspection, the supplier shall randomly select the number of samples in table II that is applicable to the quantity of items being purchased. Each sample item shall be inspected to determine conformance with the examination, the test, and the inspection. Should any of the sample items fail to meet either the examination or inspection below, the entire lot shall be considered defective and unsuitable for acceptance. A lot is defined as all items presented at one time for acceptance.

TABLE II. Sampling plan.

Quantity of items being purchased	Sample size	Quantity of items being purchased	Sample size
2 to 8	2	151 to 280	32
9 to 15	3	281 to 500	50
16 to 25	5	501 to 1,200	80
26 to 50	8	1201 to 3,200	125
51 to 90	13	3201 to 10,000	200
91 to 150	20		

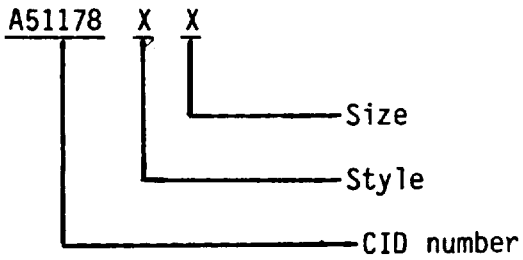
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:52Z
Check the source to verify that this is the current version before use.

Examination . Each anode submitted for sample inspection shall be visually examined to determine conformance with all requirements of this description.

Chemical composition. One anode from each lot presented for acceptance shall be tested and analyzed to determine conformance with table I. Chemical element composition of the anodes shall be determined by standard methods of chemical analysis procedures. If the anode does not conform to the chemical composition of table I, the entire lot shall be rejected.

Preservation, packaging, packing, and marking. Unless otherwise specified in the contract or purchase order preservation, packaging, packing, and marking shall be in accordance with ASTM D 3951. When specified, preservation, packaging, packing, and marking shall be in accordance with MIL-W-45562, level as specified in the contract.

CID based part identification number. The following part identification numbering procedure is for Government purposes and does not constitute a requirement for the contractor. Where applicable Arabic numbers shall be substituted for Roman numerals.



Notes.

Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

a. Title, number, and date of this commercial item description.

b. Style, size, and quantity required.

Federal Standard FED-STD-376 is available from the Naval Publications and Form center, 5801 Tabor Avenue, Philadelphia, PA 19120-5099.

Military Specification MIL-W-45562 is available from the Naval Publications and Form center, 5801 Tabor Avenue, Philadelphia, PA 19120-5099.

ASTM D 3951 is available from American Society for Testing and Materials, 1916 Race street, Philadelphia, PA 19103-1187.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:52Z
Check the source to verify that this is the current version before use.

MILITARY INTEREST:

Custodians

Navy - AS
Air Force - 99

Review Activity

Air Force - 84
DLA - GS

User Activities

Navy - OS,SH

CIVIL AGENCY COORDINATION ACTIVITY:

GSA - FSS

PREPARING ACTIVITY:

DLA - IP

Project 3426-0045

Source: https://assist.dla.mil Downloaded: 2026-10-26T08:52Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:52Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE *(YYMMDD)*
3. DOCUMENT TITLE		
4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*		
5. REASON FOR RECOMMENDATION		

6. SUBMITTER		
a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	7. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY	
a. NAME	b. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON
c. ADDRESS *(Include Zip Code)*	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

DD Form 1426, OCT 89 *Previous editions are obsolete.* 198/290

Source: https://assist.dla.mil - Downloaded: 2026-10-26T08:52Z
Check the source to verify that this is the current version before use.